

IMGS-261 Solutions to Homework #9

1. For $f_1[x] = \text{SINC}[x] \equiv \frac{\sin[\pi x]}{\pi x}$, use the “modulation theorem” to evaluate and sketch the Fourier transform of $f_2[x] = f_1[x] \cdot f_1[x] = (f_1[x])^2$

Solution: *We know that*

$$\mathcal{F}\{\text{RECT}[x]\} = \text{SINC}[\xi]$$

so we use the “transform-of-a-transform” theorem to show that:

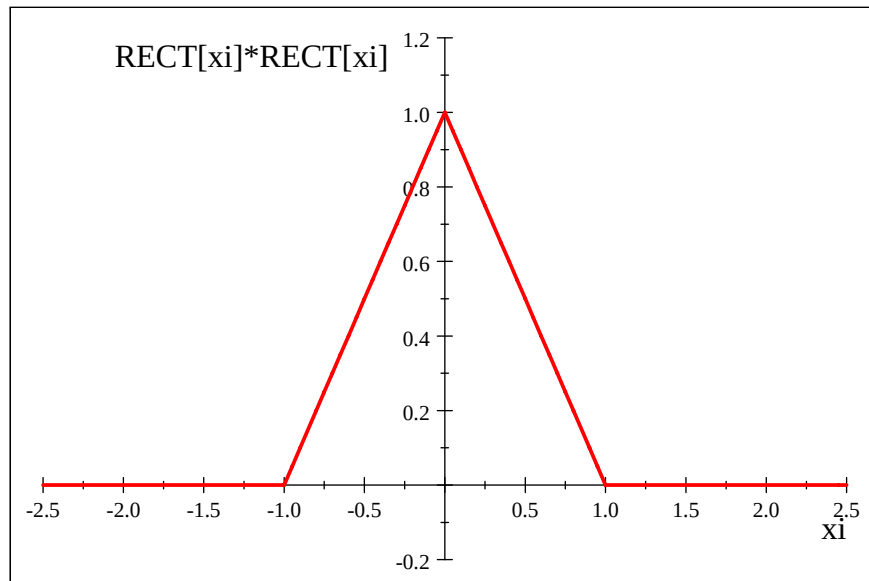
$$\mathcal{F}\{\text{SINC}[x]\} = \text{RECT}[-\xi] = \text{RECT}[+\xi] = \text{RECT}[\xi]$$

because rectangle is even function. The modulation theorem says that:

$$\begin{aligned} \mathcal{F}\{f[x] \cdot m[x]\} &= F[\xi] * M[\xi] \\ \implies \mathcal{F}\{\text{SINC}^2[x]\} &= \mathcal{F}\{\text{SINC}[x] \cdot \text{SINC}[x]\} = \text{RECT}[\xi] * \text{RECT}[\xi] \end{aligned}$$

We have already evaluated the convolution of the rectangle with itself:

$$\begin{aligned} \text{RECT}[\xi] * \text{RECT}[\xi] &= \int_{\beta=-\infty}^{\beta=+\infty} \text{RECT}[\xi - \beta] \cdot \text{RECT}[\beta] \, d\beta \\ &= \int_{\beta=-\frac{1}{2}}^{\beta=x+\frac{1}{2}} 1 \cdot 1 \, d\beta \text{ for } -1 \leq x \leq 0 \\ &= \begin{cases} 0 & \text{if } \xi > +1 \\ \int_{\beta=\xi-\frac{1}{2}}^{\beta=+\frac{1}{2}} 1 \cdot 1 \, d\beta & \text{if } 0 \leq \xi \leq +1 \\ \int_{\beta=-\frac{1}{2}}^{\beta=\xi+\frac{1}{2}} 1 \cdot 1 \, d\beta & \text{if } -1 \leq \xi \leq 0 \\ 0 & \text{if } \xi < -1 \end{cases} \\ &= \begin{cases} 0 & \text{if } \xi > +1 \\ \left(+\frac{1}{2}\right) - \left(\xi - \frac{1}{2}\right) & \text{if } 0 \leq \xi \leq +1 \\ \left(\xi + \frac{1}{2}\right) - \left(-\frac{1}{2}\right) & \text{if } -1 \leq \xi \leq 0 \\ 0 & \text{if } \xi < -1 \end{cases} \\ &= \begin{cases} 0 & \text{if } \xi > +1 \\ 1 - \xi & \text{if } 0 \leq \xi \leq +1 \\ 1 + \xi & \text{if } -1 \leq \xi \leq 0 \\ 0 & \text{if } \xi < -1 \end{cases} \\ &= (1 - |\xi|) \cdot \text{RECT}\left[\frac{\xi}{2}\right] \end{aligned}$$



*convolution of $\text{RECT} [\xi] * \text{RECT} [\xi] = \text{TRI} [\xi]$*

Note that the convolution of the rectangle with itself is a function with area equal to the product of the areas of the two component functions ($1 \cdot 1 = 1$) and support equal to the sum of the supports: $(1 + 1)$. The two components of the piecewise result are linear functions of ξ .

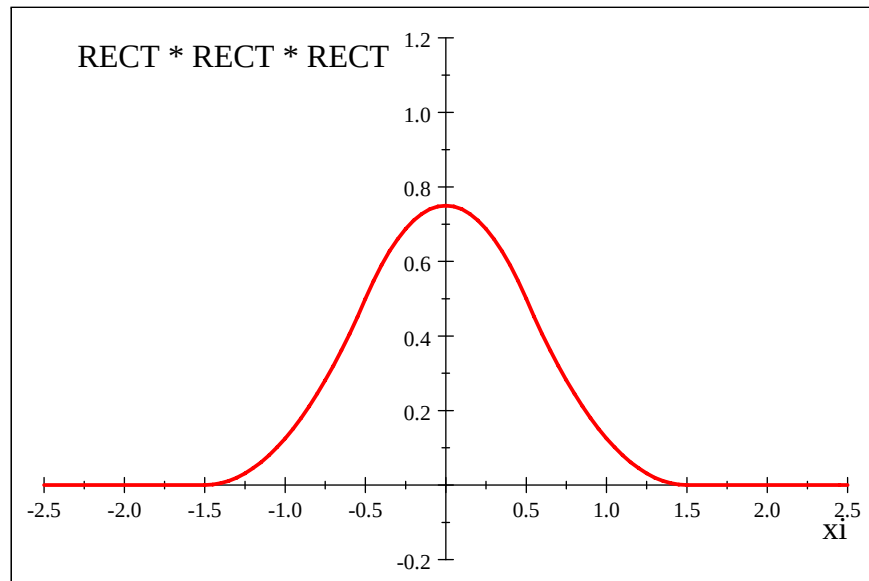
2. Use the result of #1 to sketch the form of the Fourier transform of $f_3[x] = f_1[x] \cdot f_1[x] \cdot f_1[x] = (f_1[x])^3$; you may evaluate the functional form first, but you only “have” to sketch the result.

Solution:

$$\begin{aligned}\mathcal{F}\{SINC^3[x]\} &= \mathcal{F}\{SINC[x] \cdot SINC^2[x]\} \\ &= RECT[\xi] * TRI[\xi]\end{aligned}$$

This is the convolution of a triangle and a rectangle; you only have to sketch it. The result is a function with support of 3 units ($1+1+1=3$) and area of 1 ($1 \cdot 1 \cdot 1 = 1$). The result has a “smaller” amplitude at the origin. It is easy to show that the amplitude of the convolution at the origin is the area of the triangle truncated to unit width, which is $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$. You did not have to evaluate the convolution, but I did it below for information only:

$$\begin{aligned}RECT[\xi] * TRI[\xi] &= \int_{\beta=-\infty}^{\beta=+\infty} RECT[\xi - \beta] \cdot TRI[\beta] d\beta \\ &= \int_{\beta=-1}^{\beta=\xi+\frac{1}{2}} 1 \cdot (1 + \beta) d\beta \\ &= \begin{cases} 0 & \text{if } \xi > \frac{3}{2} \\ \int_{\xi-\frac{1}{2}}^{+1} 1 \cdot (1 - \beta) d\beta & \text{if } \frac{1}{2} < \xi < \frac{3}{2} \\ \int_{\xi-\frac{1}{2}}^0 1 \cdot (1 + \beta) d\beta + \int_0^{\xi+\frac{1}{2}} 1 \cdot (1 - \beta) d\beta & \text{if } -\frac{1}{2} < \xi < \frac{1}{2} \\ \int_{-1}^{\xi+\frac{1}{2}} 1 \cdot (1 + \beta) d\beta & \text{if } -\frac{3}{2} < \xi < -\frac{1}{2} \\ 0 & \text{if } \xi < -\frac{3}{2} \end{cases} \\ &= \begin{cases} 0 & \text{if } \xi > \frac{3}{2} \\ \frac{1}{8}(2\xi - 3)^2 & \text{if } \frac{1}{2} < \xi < \frac{3}{2} \\ \frac{3}{4} - \xi^2 & \text{if } -\frac{1}{2} < \xi < \frac{1}{2} \\ \frac{1}{8}(2\xi + 3)^2 & \text{if } -\frac{3}{2} < \xi < -\frac{1}{2} \\ 0 & \text{if } \xi < -\frac{3}{2} \end{cases}\end{aligned}$$

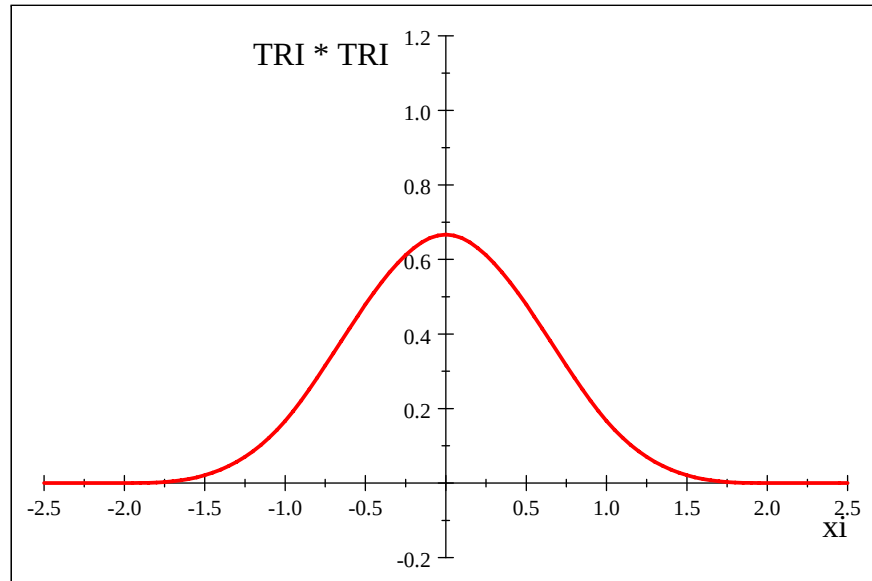


Note that each component part of the piecewise function is a quadratic function of ξ .

3. Use the result of #2 to sketch the form of the Fourier transform of $f_3[x] = f_1[x] \cdot f_1[x] \cdot f_1[x] = (f_1[x])^4$.

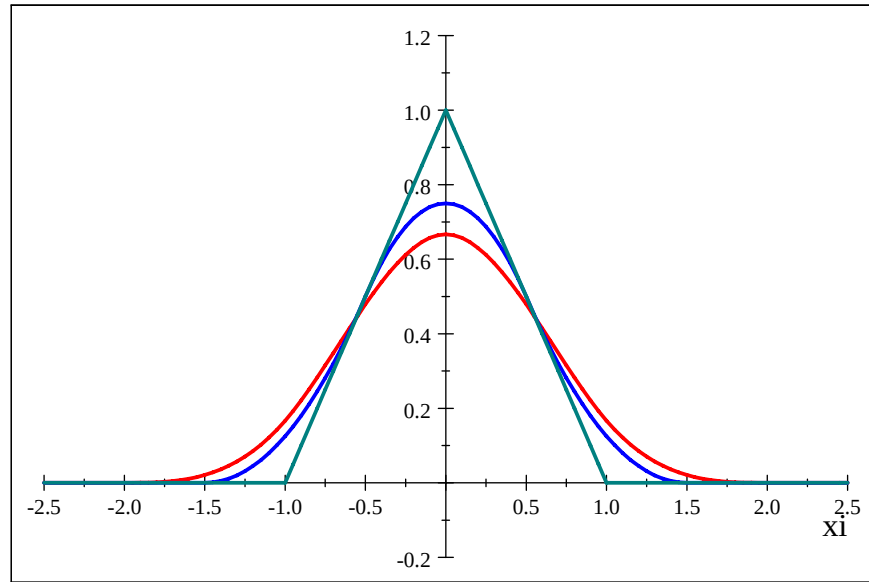
This is the convolution of the triangle with itself or of the rectangle with the result of #2. Again, the area of the convolution is the product of the areas of the component functions ($1 \cdot 1 \cdot 1 \cdot 1 = 1$) and the support is the sum of the supports ($1 + 1 + 1 + 1 = 4$). Again, you only had to sketch it, but I evaluated it for exercise:

$$\begin{aligned}
 & \text{RECT}[\xi] * (\text{RECT}[\xi] * \text{RECT}[\xi] * \text{RECT}[\xi]) \\
 &= \int_{\beta=-\infty}^{\beta=+\infty} \text{RECT}[\xi - \beta] \cdot \begin{cases} 0 & \text{if } \beta > \frac{3}{2} \\ \frac{1}{8}(2\beta - 3)^2 & \text{if } \frac{1}{2} < \beta < \frac{3}{2} \\ \frac{3}{4} - \beta^2 & \text{if } -\frac{1}{2} < \beta < \frac{1}{2} \\ \frac{1}{8}(2\beta + 3)^2 & \text{if } -\frac{3}{2} < \beta < -\frac{1}{2} \\ 0 & \text{if } \beta < -\frac{3}{2} \end{cases} d\beta \\
 &= \begin{cases} 0 & \text{if } \xi > 2 \\ \int_{\xi-\frac{3}{2}}^{\frac{3}{2}} \frac{1}{8}(2\beta - 3)^2 d\beta & \text{if } 1 < \xi < 2 \\ \int_{\xi-\frac{1}{2}}^{\frac{1}{2}} (\frac{3}{4} - \beta^2) d\beta + \int_{\frac{1}{2}}^{\xi+\frac{1}{2}} \frac{1}{8}(2\beta - 3)^2 d\beta & \text{if } 0 < \xi < 1 \\ \int_{\xi-\frac{1}{2}}^{-\frac{1}{2}} \frac{1}{8}(2\beta + 3)^2 d\beta + \int_{-\frac{1}{2}}^{\xi+\frac{1}{2}} (\frac{3}{4} - \beta^2) d\beta & \text{if } -1 < \xi < 0 \\ \int_{-\frac{3}{2}}^{\xi+\frac{1}{2}} \frac{1}{8}(2\beta + 3)^2 d\beta & \text{if } -2 < \xi < -1 \\ 0 & \text{if } \xi < -2 \end{cases} \\
 &= \begin{cases} 0 & \text{if } \xi > 2 \\ -\frac{1}{6}(\xi - 2)^3 & \text{if } 1 < \xi < 2 \\ \frac{1}{2}\xi^3 - \xi^2 + \frac{2}{3} & \text{if } 0 < \xi < 1 \\ -\frac{1}{2}\xi^3 - \xi^2 + \frac{2}{3} & \text{if } -1 < \xi < 0 \\ \frac{1}{6}(\xi + 2)^3 & \text{if } -2 < \xi < -1 \\ 0 & \text{if } \xi < -2 \end{cases}
 \end{aligned}$$



Note that each component part of the piecewise function is a cubic function of ξ and that the maximum occurs at $\xi = 0$ with amplitude $\frac{2}{3}$

Just for fun, put all three on the same plot for comparison:



$\text{RECT}[\xi] * \text{RECT}[\xi]$ in green, $\text{RECT}[\xi] * \text{RECT}[\xi] * \text{RECT}[\xi]$ in blue, and $\text{RECT}[\xi] * \text{RECT}[\xi] * \text{RECT}[\xi] * \text{RECT}[\xi]$ in red.

4. Determine the conditions that must be satisfied for the following expression to be valid:

$$\cos [2\pi\xi_0x] * SINC \left[\frac{x}{b_0} \right] = A_0 \cos [2\pi\xi_0x]$$

Solution: Use the filter theorem:

$$\mathcal{F} \{ f [x] * h [x] \} = F [\xi] \cdot H [\xi]$$

Known transforms:

$$\begin{aligned} \mathcal{F} \{ \cos [2\pi\xi_0x] \} &= \frac{1}{2} \delta [\xi + \xi_0] + \frac{1}{2} \delta [\xi - \xi_0] \\ \mathcal{F} \{ SINC [x] \} &= RECT [\xi] \end{aligned}$$

and the scaling theorem:

$$\begin{aligned} \mathcal{F} \left\{ f \left[\frac{x}{b_0} \right] \right\} &= |b_0| \cdot F [b_0 \cdot \xi] = |b_0| \cdot F \left[\frac{\xi}{b_0^{-1}} \right] \\ \Rightarrow \mathcal{F} \left\{ SINC \left[\frac{x}{b_0} \right] \right\} &= |b_0| \cdot RECT \left[\frac{\xi}{b_0^{-1}} \right] \end{aligned}$$

$$\begin{aligned} G [\xi] &= \left(\frac{1}{2} \delta [\xi + \xi_0] + \frac{1}{2} \delta [\xi - \xi_0] \right) \cdot |b_0| \cdot RECT \left[\frac{\xi}{b_0^{-1}} \right] \\ &= \frac{|b_0|}{2} \cdot (\delta [\xi + \xi_0] + \delta [\xi - \xi_0]) \cdot RECT \left[\frac{\xi}{b_0^{-1}} \right] \end{aligned}$$

This is the product of a pair of Dirac delta functions and the rectangle.

The output function is a cosine with amplitude A_0 , so the output spectrum is:

$$\begin{aligned} G [\xi] &= \mathcal{F} \{ g [x] \} = \mathcal{F} \{ A_0 \cos [2\pi\xi_0x] \} \\ &= A_0 \cdot \mathcal{F} \{ \cos [2\pi\xi_0x] \} \\ &= \frac{A_0}{2} \cdot \delta [\xi + \xi_0] + \frac{A_0}{2} \cdot \delta [\xi - \xi_0] \end{aligned}$$

The question is whether these two expressions are equal:

$$\frac{|b_0|}{2} \cdot (\delta [\xi + \xi_0] + \delta [\xi - \xi_0]) \cdot RECT \left[\frac{\xi}{b_0^{-1}} \right] \stackrel{?}{=} \frac{A_0}{2} \cdot \delta [\xi + \xi_0] + \frac{A_0}{2} \cdot \delta [\xi - \xi_0]$$

This obviously is correct IF the frequency-domain rectangle is sufficiently wide to enclose both Dirac delta functions, i.e.,

$$\begin{aligned} b_0^{-1} &= \frac{1}{b_0} > \xi_0 - (-\xi_0) = 2\xi_0 \\ \Rightarrow \xi_0 &< \frac{1}{2b_0} \end{aligned}$$

also note that $A_0 = |b_0|$.

5. Determine the conditions that must be satisfied for the following expression to be valid:

$$\cos [2\pi\xi_0x] * \text{SINC}^2 \left[\frac{x}{b_0} \right] = B_0 \cos [2\pi\xi_0x]$$

Solution: *Again, evaluate the spectra:*

$$\begin{aligned} \mathcal{F} \{ \cos [2\pi\xi_0x] \} &= \frac{1}{2} \delta [\xi + \xi_0] + \frac{1}{2} \delta [\xi - \xi_0] \\ \mathcal{F} \left\{ \text{SINC}^2 \left[\frac{x}{b_0} \right] \right\} &= |b_0| \cdot \text{TRI} \left[\frac{\xi}{b_0^{-1}} \right] \\ \mathcal{F} \{ B_0 \cos [2\pi\xi_0x] \} &= \frac{B_0}{2} \delta [\xi + \xi_0] + \frac{B_0}{2} \delta [\xi - \xi_0] \\ \mathcal{F} \left\{ \cos [2\pi\xi_0x] * \text{SINC}^2 \left[\frac{x}{b_0} \right] \right\} &= \left(\frac{1}{2} \delta [\xi + \xi_0] + \frac{1}{2} \delta [\xi - \xi_0] \right) \cdot \left(|b_0| \cdot \text{TRI} \left[\frac{\xi}{b_0^{-1}} \right] \right) \\ &= \left(\frac{|b_0|}{2} \delta [\xi + \xi_0] + \frac{|b_0|}{2} \delta [\xi - \xi_0] \right) \cdot \text{TRI} \left[\frac{\xi}{b_0^{-1}} \right] \end{aligned}$$

The convolution with $\text{SINC}^2 \left[\frac{x}{b_0} \right]$ will multiply the spectra by the triangle, so the output will be a cosine with a reduced amplitude. The definition of the Triangle is:

$$\begin{aligned} \text{TRI} [x] &= \begin{cases} 0 & \text{if } x > 1 \\ 1 - x & \text{if } 0 \leq x \leq 1 \\ 1 + x & \text{if } -1 \leq x \leq 0 \\ 0 & \text{if } x < -1 \end{cases} \\ \Rightarrow \text{TRI} \left[\frac{\xi}{b_0^{-1}} \right] &= \text{TRI} [b_0\xi] = \begin{cases} 0 & \text{if } b_0\xi > 1 \\ 1 - b_0\xi & \text{if } 0 \leq b_0\xi \leq 1 \\ 1 + b_0\xi & \text{if } -1 \leq b_0\xi \leq 0 \\ 0 & \text{if } b_0\xi < -1 \end{cases} \\ \Rightarrow &\begin{cases} 0 & \text{if } \xi > \frac{1}{b_0} \\ 1 - b_0\xi & \text{if } 0 \leq \xi \leq \frac{1}{b_0} \\ 1 + b_0\xi & \text{if } -\frac{1}{b_0} \leq \xi \leq 0 \\ 0 & \text{if } \xi < -\frac{1}{b_0} \end{cases} \end{aligned}$$

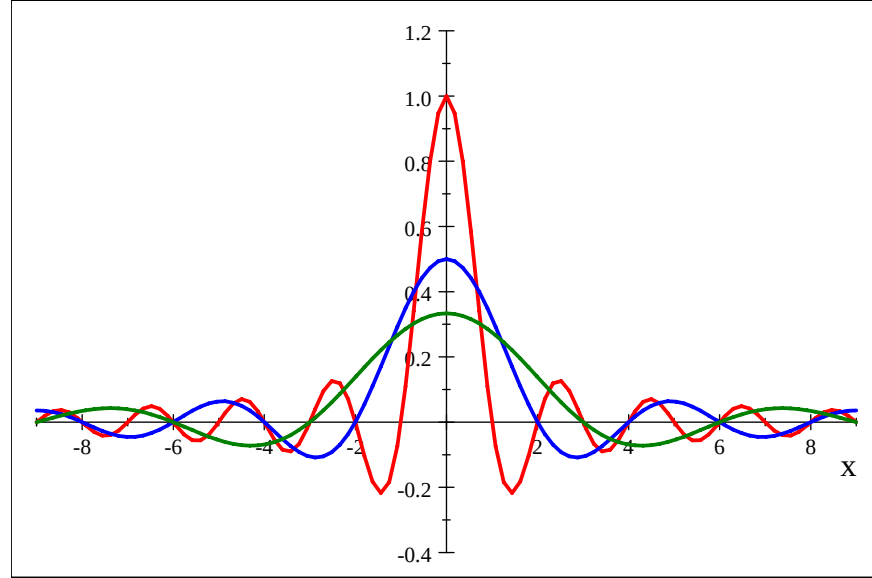
so use the property of the Dirac delta function in products:

$$\begin{aligned} f[x] \cdot \delta [x - x_0] &= f[x_0] \cdot \delta [x - x_0] \\ \left(\frac{|b_0|}{2} \delta [\xi + \xi_0] + \frac{|b_0|}{2} \delta [\xi - \xi_0] \right) \cdot \text{TRI} \left[\frac{\xi}{b_0^{-1}} \right] &= \frac{|b_0|}{2} (\delta [\xi + \xi_0] + \delta [\xi - \xi_0]) \cdot \left(1 - \frac{|\xi_0|}{b_0} \right) \\ &= \left(|b_0| \cdot \left(1 - \frac{|\xi_0|}{b_0} \right) \right) \cdot \frac{1}{2} (\delta [\xi + \xi_0] + \delta [\xi - \xi_0]) \\ B_0 &= \begin{cases} 0 & \text{if } |\xi_0| > b_0 \\ |b_0| \cdot (1 - |\xi_0|) & \text{if } |\xi_0| < b_0 \end{cases} \end{aligned}$$

6. Consider the convolution of 100 scaled *SINC* functions with width parameters that “start small” and grow for succeeding terms:

$$g[x] = \frac{1}{1} \text{SINC} \left[\frac{x}{1} \right] * \frac{1}{2} \text{SINC} \left[\frac{x}{2} \right] * \frac{1}{3} \text{SINC} \left[\frac{x}{3} \right] * \cdots * \frac{1}{n} \text{SINC} \left[\frac{x}{n} \right] * \cdots * \frac{1}{100} \text{SINC} \left[\frac{x}{100} \right]$$

- (a) Sketch the first three *SINC* functions on the same graph, including both the amplitudes and “width” parameters.



(red) $\text{SINC}[x]$, (blue) $\frac{1}{2} \text{SINC} \left[\frac{x}{2} \right]$, (green) $\frac{1}{3} \text{SINC} \left[\frac{x}{3} \right]$

- (b) Evaluate the functional form of and sketch $g[x]$.

Solution: use the filter theorem:

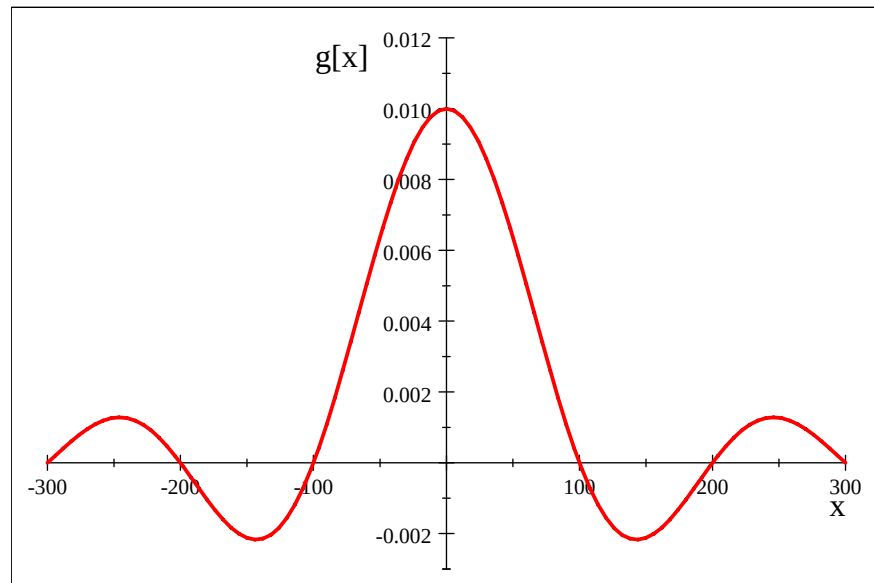
$$\begin{aligned} G[\xi] &= \mathcal{F} \left\{ \frac{1}{1} \text{SINC} \left[\frac{x}{1} \right] \right\} \cdot \mathcal{F} \left\{ \frac{1}{2} \text{SINC} \left[\frac{x}{2} \right] \right\} \cdots \mathcal{F} \left\{ \frac{1}{100} \text{SINC} \left[\frac{x}{100} \right] \right\} \\ &= \text{RECT}[\xi] \cdot \text{RECT} \left[\frac{\xi}{\left(\frac{1}{2} \right)} \right] \cdot \text{RECT} \left[\frac{\xi}{\left(\frac{1}{3} \right)} \right] \cdots \text{RECT} \left[\frac{\xi}{\left(\frac{1}{100} \right)} \right] \end{aligned}$$

The product of these rectangles with different widths and the same unit height is the single narrowest rectangle:

$$G[\xi] = \text{RECT} \left[\frac{\xi}{\left(\frac{1}{100} \right)} \right]$$

Now evaluate the inverse Fourier transform to see that the result is the “widest” *SINC* function

$$g[x] = \frac{1}{100} \cdot \text{SINC} \left[\frac{x}{100} \right]$$



$$g[x] = \frac{1}{1} \text{SINC} \left[\frac{x}{1} \right] * \frac{1}{2} \text{SINC} \left[\frac{x}{2} \right] * \cdots * \frac{1}{n} \text{SINC} \left[\frac{x}{n} \right] * \cdots * \frac{1}{100} \text{SINC} \left[\frac{x}{100} \right]$$

7. Consider the convolution of a $SINC$ function with width b_0 and a $SINC^2$ function with width d_0 :

$$SINC \left[\frac{x}{b_0} \right] * SINC^2 \left[\frac{x}{d_0} \right] = g[x] = A_0 \cdot SINC^2 \left[\frac{x}{d_0} \right]$$

- (a) Find the conditions on the width parameters b_0 , d_0 , and A_0 that must be satisfied for this expression to be valid.

Solution: Evaluate in the frequency domain using the filter theorem

$$\begin{aligned} \mathcal{F} \left\{ SINC \left[\frac{x}{b_0} \right] * SINC^2 \left[\frac{x}{d_0} \right] \right\} &= \mathcal{F} \left\{ SINC \left[\frac{x}{b_0} \right] \right\} \cdot \mathcal{F} \left\{ SINC^2 \left[\frac{x}{d_0} \right] \right\} \\ &= |b_0| \cdot RECT \left[\frac{\xi}{b_0^{-1}} \right] \cdot |d_0| \cdot TRI \left[\frac{\xi}{d_0^{-1}} \right] \\ &= |b_0| \cdot |d_0| \cdot RECT \left[\frac{\xi}{b_0^{-1}} \right] \cdot TRI \left[\frac{\xi}{d_0^{-1}} \right] \end{aligned}$$

The output spectrum is:

$$\begin{aligned} G[\xi] &= \mathcal{F} \left\{ A_0 \cdot SINC^2 \left[\frac{x}{d_0} \right] \right\} \\ &= A_0 \cdot \mathcal{F} \left\{ SINC^2 \left[\frac{x}{d_0} \right] \right\} \\ &= A_0 \cdot TRI \left[\frac{\xi}{d_0^{-1}} \right] \end{aligned}$$

The product of the frequency-domain rectangle and triangle must be the triangle, which means that the width parameters must be related:

$$\frac{1}{2}b_0^{-1} \geq d_0^{-1} \implies \frac{1}{2b_0} \geq \frac{1}{d_0} \implies \boxed{\frac{d_0}{b_0} \geq 2}$$

Under these conditions, the output spectrum is:

$$\begin{aligned} G[\xi] &= |b_0| \cdot |d_0| \cdot RECT \left[\frac{\xi}{b_0^{-1}} \right] \cdot TRI \left[\frac{\xi}{d_0^{-1}} \right] \\ &= |b_0| \cdot |d_0| \cdot TRI \left[\frac{\xi}{d_0^{-1}} \right] \\ &= |b_0| \cdot SINC^2 \left[\frac{x}{d_0} \right] \\ \implies A_0 &= |b_0| \end{aligned}$$

- (b) Explain whether it is possible to select width parameters and B_0 such that:

$$SINC \left[\frac{x}{b_0} \right] * SINC^2 \left[\frac{x}{d_0} \right] = g[x] = B_0 \cdot SINC \left[\frac{x}{b_0} \right]$$

In this case, the frequency-domain product of the rectangle and the triangle must be a rectangle; the product of the rectangle and the triangle is a windowed triangle, and NOT a rectangle, so this cannot be done.